

Session 16 Quiz - Voice AI

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID : 53624]

_____ (also called STT) is the Voice AI technology that converts spoken audio into written text - turning a waveform of your voice into the words you said.

- A. TTS (Text-to-Speech)
- B. ASR (Automatic Speech Recognition)
- C. NLU (Natural Language Understanding)
- D. OCR (Optical Character Recognition)

Correct Answer: B

Explanation: ASR (also called STT) is the "ear" of Voice AI. It converts the waveform captured by a microphone into the text the system can then process. TTS goes the opposite direction (text → speech), NLU figures out meaning from text, and OCR is for reading text from images.

Q2 [Type: MCQ | Difficulty: Easy | QID : 53626]

_____ is the Voice AI technology that converts written text into spoken audio - for example, narrating an audiobook in a natural-sounding voice.

- A. ASR (Automatic Speech Recognition)
- B. STT (Speech-to-Text)
- C. TTS (Text-to-Speech)
- D. NLU (Natural Language Understanding)

Correct Answer: C

Explanation: TTS is the "mouth" of Voice AI. Modern TTS systems like ElevenLabs use deep learning to predict natural prosody - making the synthetic voice sound expressive rather than robotic.

Q3 [Type: MCQ | Difficulty: Medium | QID : 53627]

_____ is the rhythm, pitch, stress, and emotion of speech - the "music" of how words are said - which modern TTS systems use to make AI voices sound human rather than robotic.

- A. Prosody
- B. Phoneme
- C. Waveform
- D. Sampling rate

Correct Answer: A

Explanation: Prosody is what makes AI voices sound natural - where to pause, which words to emphasise, what tone to use. Without it, speech is flat and robotic; with it, modern TTS sounds remarkably human.

Q4 [Type: MCQ | Difficulty: Easy | QID : 53628]

Voice AI systems truly hear and understand spoken language the same way a human ear and brain do.

- A. True
- B. False

Correct Answer: B (False)

Explanation: Voice AI finds mathematical patterns in sound waves and matches them to words it has seen millions of times. The AI does not "know" what a reminder is or what "home" means - it has learned that the words "remind me to [task] at [time]" usually mean schedule a notification. This is pattern matching, not comprehension, and it parallels the same warning we saw in NLP and Computer Vision.

Q5 [Type: MCQ | Difficulty: Medium | QID : 53629]

Modern ASR/STT systems perform equally well across all accents, dialects, and demographic groups.

- A. True
- B. False

Correct Answer: B (False)

Explanation: Both Accents and Bias are named challenges in Voice AI. ASR systems often have noticeably higher error rates for women and people of colour when training data underrepresents those groups, and they can fail on regional accents like Scottish or rural Irish that differ from what they were trained on. Error rates can be 20% higher for female voices than male voices if training data was not balanced.

Q6 [Type: MCQ | Difficulty: Medium | QID : 53630]

Prosody - the rhythm, pitch, and stress of speech - carries meaning and emotion, so AI voices without it sound flat and robotic.

- A. True
- B. False

Correct Answer: A (True)

Explanation: Prosody is what makes AI voices sound human. Without it, speech would be flat and robotic - it is the layer that tells the system where to pause, which words to emphasise, and what emotion to convey. Try comparing a flat read of "I can't believe you did that" with a surprised, emphasised version. Same words, very different meaning. That gap is prosody - and it is the reason modern TTS sounds natural while older systems sounded like robots.

Q7 [Type: MCQ | Difficulty: Easy | QID : 53632]

What is the correct order of the three stages of the Voice AI framework?

- A. Speak → Hear → Process
- B. Process → Hear → Speak
- C. Hear → Process → Speak
- D. Listen → Translate → Print

Correct Answer: C

Explanation: Hear (capture sound as a waveform) → Process (ASR/STT → NLU → Response Planning) → Speak (TTS converts the response into audio). This matches the Sense → Think → Act framework introduced earlier in the course.

Q8 [Type: MCQ | Difficulty: Easy | QID : 53633]

What is a "phoneme" in Voice AI?

- A. A wiggly graph showing sound over time
- B. The rhythm and stress of a sentence
- C. A way to compress audio files
- D. The smallest unit of sound in a language

Correct Answer: D

Explanation: Phonemes are the building blocks of spoken language. English has about 40 phonemes, and ASR/STT works by combining acoustic features into phonemes and then assembling those phonemes into words using a dictionary and context.

Q9 [Type: MCQ | Difficulty: Medium | QID : 53636]

Inside the Process stage, ASR/STT follows a specific four-step flow. What is the correct order?

- A. Words → Phonemes → Acoustic Features → Sound Waves
- B. Phonemes → Sound Waves → Words → Acoustic Features
- C. Sound Waves → Acoustic Features → Phonemes → Words
- D. Acoustic Features → Sound Waves → Words → Phonemes

Correct Answer: C

Explanation: Each step builds on the previous one: raw sound is captured, then features (pitch, energy, frequency) are extracted, then those features are mapped into the language's phonemes, and finally the phonemes are assembled into words using a dictionary and context.

Q10 [Type: MCQ | Difficulty: Medium | QID : 53637]

When a user says "Remind me to call home at 6 PM," identifying "6 PM" as the time is an example of identifying which NLU element?

- A. Sentiment
- B. Phonemes
- C. Intent
- D. Entities

Correct Answer: D

Explanation: Entities are the specific details or parameters within a command - times, locations, names, durations. Intent is the action the user wants (set_reminder); entities are the supporting details that make the action concrete.

Q11 [Type: MCQ | Difficulty: Hard | QID : 53638]

A publisher wants to generate an entire audiobook in a warm, friendly narrator voice - starting from the printed text of the book. Which Voice AI technology is the main one needed for this job?

- A. ASR/STT, to convert spoken sound recordings into written text for the audiobook
- B. TTS, to convert written text into a natural-sounding voice with appropriate prosody
- C. Speaker recognition, to identify who is reading the book aloud during recording
- D. NLU, to figure out what the listener wants to hear from the audiobook

Correct Answer: B

Explanation: The job is generating spoken audio from text - exactly what TTS is built for. Modern TTS systems use prosody prediction to make the narration sound human rather than robotic. ASR/STT goes the opposite direction (speech → text); speaker recognition identifies voices; NLU is about understanding intent.

Q12 [Type: MCQ | Difficulty: Medium | QID : 53639]

A key idea about Voice AI is that it is "pattern matching, not true understanding." Which example BEST illustrates this distinction?

- A. "Remind me to call home at 6 PM," the AI sets the reminder but does not know what "home" means
- B. Voice assistants respond to user commands in less than a second, making them feel highly responsive
- C. The system uses a microphone to capture sound waves and convert them into a digital signal
- D. ElevenLabs lets you create custom voices that can sound like specific people or characters

Correct Answer: A

Explanation: The system correctly executes the command not because it conceptually understands "home" or "reminder," but because it matched the audio + text pattern to learned examples. Pattern matching can succeed without comprehension - and it fails when the pattern shifts (accent, noise, sarcasm).

Q13 [Type: MCQ | Difficulty: Hard | QID : 53640]

A nonprofit serving adults with speech disabilities is evaluating two Voice AI systems. System A has higher overall accuracy but was tested only on neurotypical speech. System B has slightly lower overall accuracy but was trained on diverse speech patterns including dysarthria and stuttering, and the nonprofit's users perform significantly better on it in testing. Which system should the nonprofit choose and why?

- A.** System A - overall accuracy is the most reliable indicator of how well a Voice AI system will perform in real-world use
- B.** System A - higher accuracy will improve over time and close the gap for users with speech disabilities as the technology matures
- C.** System B - what matters is how well it serves these specific users, and testing shows it performs better for them
- D.** Neither system - the nonprofit should wait until accuracy is high across all speech patterns before deploying any Voice AI tool

Correct Answer: C

Explanation: The system that performs better for the people actually using it is the right system - overall accuracy is only meaningful if it reflects your users, and here it does not. This is the same point that came up with CV training-deployment mismatch: a benchmark on the wrong population is not a benchmark for your population.

Q14 [Type: MCQ | Difficulty: Hard | QID : 53641]

A clinic uses Voice AI to transcribe doctor dictations. After three months, error rates are significantly higher for doctors who are non-native English speakers. The administrator suggests those doctors "speak more slowly and clearly." What is the most important problem with this suggestion?

- A.** Speaking more slowly would actually confuse the AI further - transcription models perform worse with artificially slow input and unnatural pacing
- B.** The suggestion is impractical because doctors are too time-pressured during patient care to consistently adjust their speaking style for the system
- C.** Non-native speaking doctors should not use Voice AI tools at all until the underlying technology improves enough to handle diverse accents
- D.** It shifts the burden from the system to the doctors, and likely won't work since the model wasn't trained on enough non-native speech

Correct Answer: D

Explanation: The administrator has misdiagnosed the problem. When a system performs worse for a specific group, the cause is almost always in what the system was trained on, not in how that group behaves. Asking users to change to fit the system inverts the responsibility and rarely closes the gap.

Q15 [Type: MCQ | Difficulty: Medium | QID : 53642]

Which Voice AI task is specifically designed to identify WHO is speaking rather than WHAT is being said?

- A. Voice Commands
- B. Voice Cloning
- C. Speech-to-Text (STT)
- D. Speaker Recognition

Correct Answer: D

Explanation: Speaker recognition analyses voiceprints to identify an individual based on their unique vocal characteristics. STT converts speech to text (the WHAT). Voice cloning creates a synthetic copy of a specific person's voice. Voice commands are spoken instructions the system executes.

Q16 [Type: MCQ | Difficulty: Medium | QID : 53643]

How does Voice AI assist people with visual impairments (blind or low vision)?

- A. Live captioning translates spoken words into text on screen
- B. Screen readers use TTS to speak text on the screen aloud
- C. ASR/STT lets them type using their voice
- D. Audiobooks replace the need for physical braille

Correct Answer: B

Explanation: Text-to-Speech allows users who cannot see a screen to hear the information that is written on it - button labels, paragraph text, image descriptions. This is one of the longest-running accessibility applications of Voice AI. Live captioning (A) helps deaf and hard-of-hearing users, the reverse application.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 53644]

Scenario: A smart speaker in a busy kitchen hears someone say "Add milk to the shopping list," but the transcription comes out as "Add silk to the shopping list" - because pots, dishes, and conversation were happening at the same time.

Question: Which Voice AI challenge is most clearly at work?

- A. Homophones
- B. Background noise
- C. Privacy
- D. Emotional tone

Correct Answer: B

Explanation: Other sounds in the kitchen mixed with the voice, distorting the acoustic features the system uses to identify phonemes. The result is a confused mapping from sound to words - the textbook background-noise failure mode.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID : 53645]

Scenario: After a long, stressful day, a user looks at a stack of homework and says flatly, "Oh, wonderful." A sentiment-aware voice assistant analyses the speech and labels the sentiment as POSITIVE.

Question: Which Voice AI challenge is at work?

- A. Background noise (environment) - kitchen sounds in the room confused the system's audio processing
- B. Privacy (data collection) - the system should not have been listening to private moments in the home
- C. Emotional tone (sarcasm) - the words say "wonderful," but the flat delivery flips the meaning
- D. Accent (regional speech) - the user has a particular regional dialect the system was not trained on

Correct Answer: C

Explanation: The speaker means the opposite of what the words say. Voice AI patterns key off the word "wonderful" (typically positive), and even sentiment-aware systems often miss the inverting effect of flat or sarcastic delivery - connecting subtle audio cues of sarcasm to a meaning reversal is genuinely hard.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID : 53646]

Scenario: A hospital is considering deploying Voice AI-based transcription for all patient consultations. The transcription would create written records of conversations between doctors and patients automatically. The chief medical officer says: "This will let doctors focus on patients instead of typing notes."

Question: Identify one significant benefit AND one significant risk of this plan.

(Respond in 3–4 sentences.)

Model Answer: A significant benefit is that doctors can give patients their full attention during consultations rather than typing notes, which improves both the quality of the patient interaction and reduces physician burnout from documentation. A significant risk is that Voice AI transcription can mishear medical terminology, miss critical details from non-standard speech (accents, soft-spoken patients, patients with speech difficulties), and create a permanent written record of conversations patients may not realise is being made - raising both accuracy and consent concerns for sensitive medical information.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 53647]

Scenario: A small business is considering using AI voice cloning to create a synthetic version of their CEO's voice for company-wide audio announcements. This would save the CEO from recording dozens of updates manually each year. The CEO supports the idea; the head of communications worries about misuse and what happens when employees hear voice messages that are not actually from the CEO.

Question: Identify one significant benefit AND one significant risk of using voice cloning in this corporate context, and describe what the company should require before deploying it.

(Respond in 3–4 sentences.)

Model Answer: A significant benefit is that voice cloning can dramatically reduce the CEO's time spent on routine announcements while maintaining the personal feel of a CEO message - letting the CEO focus on higher-value work and the company communicate more frequently. A significant risk is that if employees cannot tell whether voice messages are genuine, the trust value of hearing the CEO directly is undermined - and worse, if the voice clone is compromised or used inappropriately, it could spread false statements that look authentically from the CEO. To deploy responsibly, the company should require that every voice-cloned message be clearly labelled as AI-generated, restrict access to the voice clone with multiple approval requirements for each use, and have the CEO approve every script before generation - treating the voice clone as a tool that needs oversight, not as a substitute for personal communication.